

Global Dental Workforce and Basic Oral-Health Service Availability

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A problem and why it matters

The World Health Organization (WHO) estimated that approximately 3.5 billion people were affected by oral diseases, with three out of four affected people living in low- and middle-income countries. The burden falls disproportionately on disadvantaged populations (WHO, 2022c). In 2022, the World Health Assembly adopted a global strategy envisioning universal health coverage for oral health by 2030 (WHO, 2022a, 2022c).

Dental care depends on trained labor. This paper asks how dentist density, measured as dentists per 10,000 people, relates to basic service availability across countries at different income levels. The economic questions concern workforce supply, training barriers, financing, and incentives to serve underserved populations.

My interpretation of the evidence

The analysis matches 171 countries using WHO dentist density, World Bank income groups, and three WHO 2021 services: screening, urgent oral care, and basic restorative care (WHO, n.d.-b; World Bank, n.d.). General availability means reaching at least half of patients in need in public primary-care facilities (WHO, 2022b, p. 2).

“Not achieved” means none of these services meets that threshold: 7/55 high-income countries (12.7%), versus 3/53 upper-middle-income countries (5.7%). It does not mean no dental care exists. Private delivery may be missed, but these data cannot establish whether it explains a country's result. The correlation concerns public-primary-care availability, not overall access.

Dentist supply and availability had a modest relationship (Spearman rank correlation 0.306). Median density was 2.97 where all three services were generally available, 0.76 where

none met the criterion, and 0.72 where one or two did. These middle values did not rise steadily across categories (Figure 1; Appendix Table A1).

Median dentist density rose with income: 0.06 in low-income, 0.31 in lower-middle-income, 2.33 in upper-middle-income, and 7.39 in high-income countries (Figure 2; Appendix Table A2). Using observations through 2021, correlations within these groups were 0.374, 0.098, -0.170, and 0.179, respectively. The upper-middle-income relationship was weak and negative; the positive relationship was largest in low-income countries. These comparisons do not isolate income's influence.

Using each country's latest dentist observation from 2021 or earlier barely changed the correlation: 0.305 versus 0.306, retaining all 171 countries. Restricting observations to 2018–2021 retained 157 countries and gave 0.325; original observations through 2023 on those same countries gave 0.340. The modest relationship persisted, although the recent-window check also excludes countries.

Older observations, today's income groups, and reporting differences remain concerns. Sources may count working or registered dentists and vary in completeness (WHO, n.d.-a). National averages hide local shortages. Income, health-system design, or spending could influence both supply and availability. These comparisons cannot establish causation or show whether financing, provider location, or sharing tasks explains the differences (Appendix Table A3).

I expected dentist scarcity to be the main explanation. The findings challenged that expectation without establishing a better explanation.

Of 171 countries, 99 answered whether essential dental treatment was included in their largest public scheme: 78 Yes and 21 No. The other 72 had no usable response and were not counted as No. Full service availability was more common among Yes respondents (Table 1).

Table 1. Essential-curative benefit inclusion and basic service availability (2021)

Benefit included in largest public scheme	Countries	All three services generally available	95% Wilson interval
Yes	78	49 (62.8%)	51.7–72.7%
No	21	9 (42.9%)	24.5–63.5%

Source: calculations from WHO 2021 benefit-package and public-primary-care service indicators (WHO, n.d.-b, n.d.-c). Intervals describe individual proportions, not their difference.

The two-sided Fisher exact test gives $p = 0.135$, without strong statistical evidence of a difference. Patterns vary by income group; some subgroups are small. Median dentist density using observations through 2021 is higher with benefit inclusion (2.23 versus 0.72). Neither the test nor intervals address income/workforce confounding or which countries reported data.

Policy options and my provisional judgment

I am considering three options:

- 1. Expand dentist training and retention.** This could build capacity but takes money and time; graduates may choose affluent or urban areas.
- 2. Improve provider location and primary-care links.** This could help reach underserved populations but requires recruitment, retention, referrals, and public-sector support.
- 3. Expand public financing for essential care.** This could improve affordability, but additional demand may exceed provider capacity. The financing comparison does not prove a provider shortage.

Recommendation. National health ministries, working with public payers and workforce planners, should match policy to capacity and patients' barriers. My findings inform this tentative judgment, without identifying a workforce threshold or best policy. Where local evidence confirms severe shortages, I would prioritize training, retention, and deployment. Where providers can deliver care but cost is the barrier, I would prioritize financing that reduces patients' costs, encourages provider participation, and supports underserved areas.

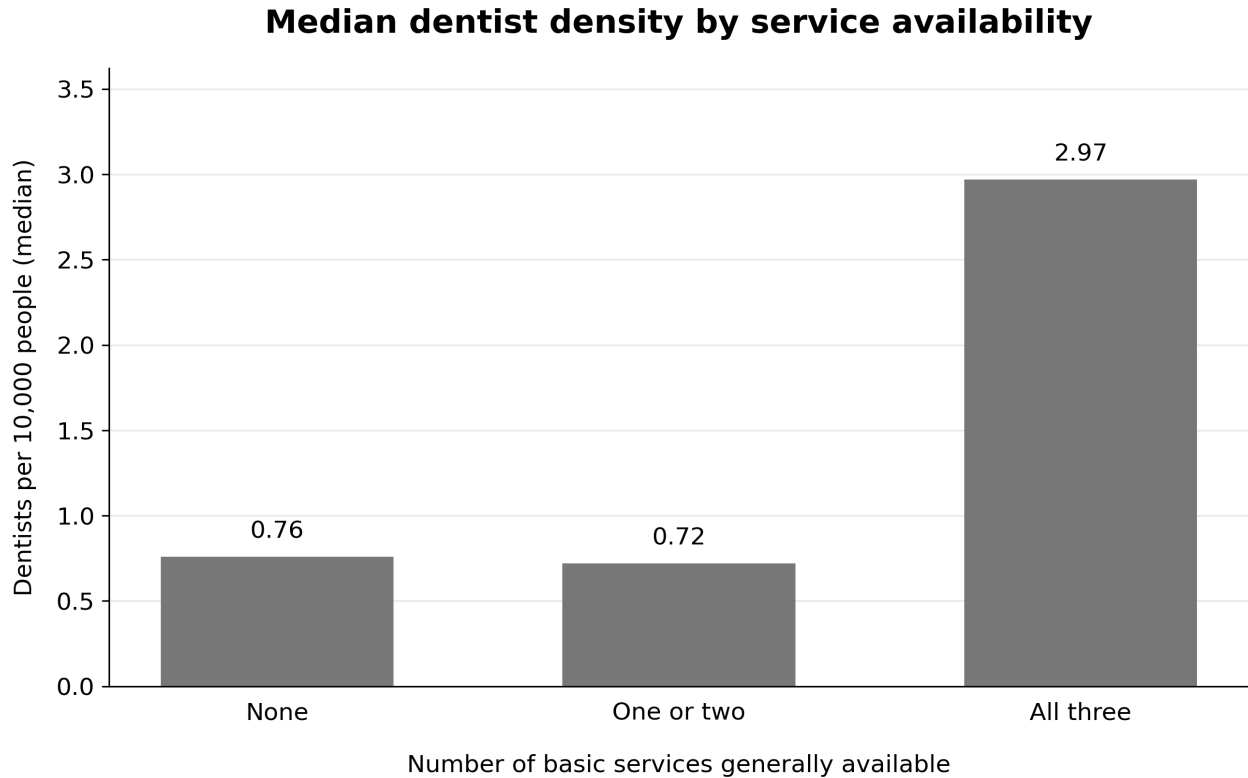
The strongest objection. Income or workforce capacity could explain both benefit inclusion and availability; these data cannot show which came first. Romania covers essential treatment, yet none of the three services is generally available in public primary-care facilities (WHO, 2022b). Only 5% of dental spending was publicly funded in 2021 (OECD/European Observatory on Health Systems and Policies, 2023, p. 15). Listing a benefit differs from reducing patients' costs. This share does not identify who receives care or measure a reform's effect; private treatment may be missed. Subsidies could raise prices or waiting times if supply cannot respond—a theoretical risk, not my finding.

My response. My findings do not prove cause and effect, so my recommendation depends on local conditions. Public payments and patients' costs should encourage affordable care and provider participation. Low payments can discourage participation, but are not established as Romania's cause. Deployment incentives may also be needed. I would pilot reforms, comparing costs with care received, affordability, unmet need, and oral health. I would reconsider if credible evaluations found no gains after benefits expanded, or better results from workforce expansion or deployment at similar supply levels.

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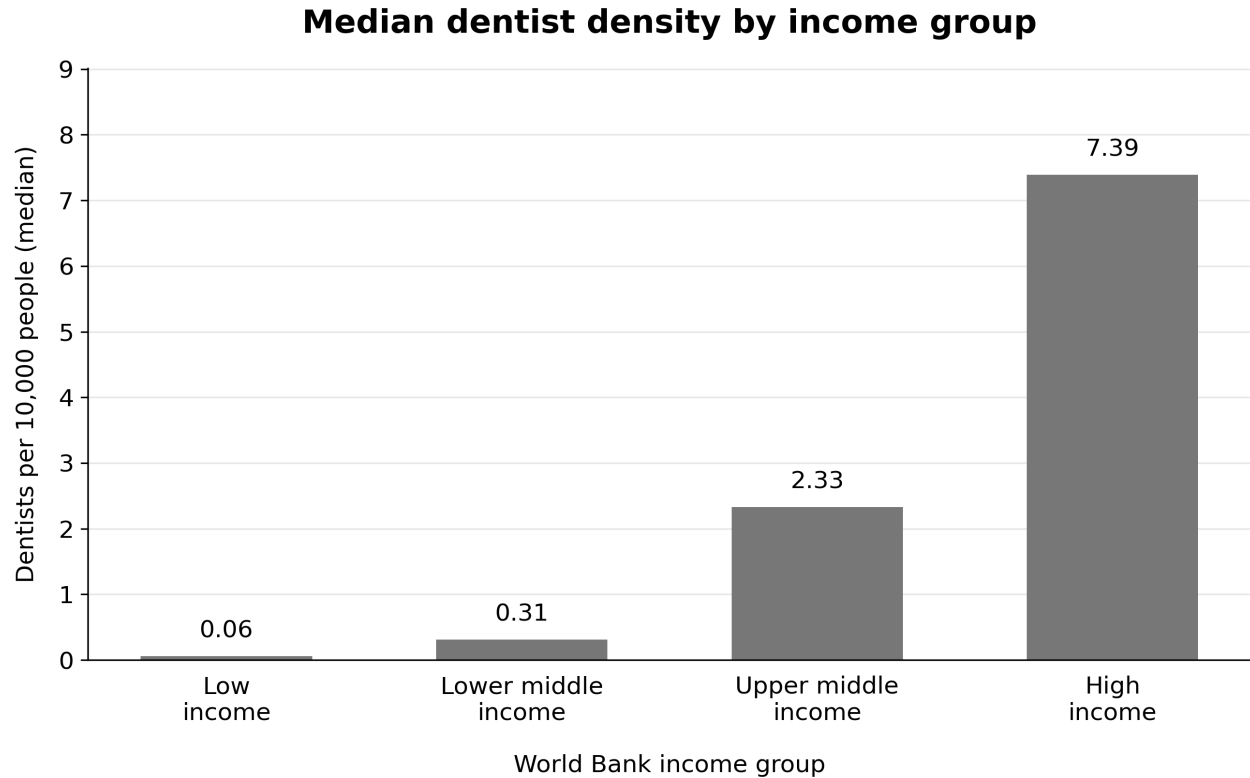
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Figure 1. Dentist density by service availability



Categories count services generally available in public primary-care facilities. Medians use workforce observations through 2023 and WHO 2021 service components. Source: author's calculations from WHO GHO/NCD CCS and World Bank data. Median comparisons do not establish causation.

Figure 2. Dentist density by income



Income groups are current World Bank API classifications, rather than historical 2021 groups.

Medians use workforce observations through 2023. Source: author's calculations from WHO workforce data and World Bank classifications.

Appendix

Appendix Table A1: Service availability and dentist density

Reported basic service availability	Median dentists per 10,000 people
None of the three services generally available	0.76
One or two services generally available	0.72
All three services generally available	2.97

Original workforce observations through 2023; outcome derived from WHO 2021 components. “Generally available” uses WHO's public-primary-care reach criterion.

Appendix Table A2: Income group and dentist density

World Bank income group	Median dentists per 10,000 people
Low income	0.06
Lower middle income	0.31
Upper middle income	2.33
High income	7.39

Source for Tables A1–A2: author's calculations from matched WHO workforce/service indicators and World Bank income classifications. These are original-baseline medians; the workforce-year sensitivity check is described in the main text.

Appendix Table A3: Limits of the analysis

What we cannot conclude	Why
More dentists cause better service availability	The analysis shows associations, not cause and effect.

What we cannot conclude	Why
Dentist supply does not matter	A weak national relationship can conceal local shortages or differences in how dentists work.
Financing, deployment, or task sharing explains the differences	These possible explanations have not yet been tested.
Patients actually receive affordable, accessible care	The outcome measures reported service availability in public primary care, not utilization, cost, travel distance, or quality.
A particular policy would improve access	The analysis has not evaluated the effects, costs, or tradeoffs of an intervention.

AI assistance disclosure

ChatGPT assisted with explanations of the analysis and policy options, table formatting, and revisions to wording and structure. Some passages originated in AI explanations shared during the drafting process. The policy preference remains provisional, and further source verification and methodological checks are pending.